GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NADH Dehydrogenase Complex Assembly (GO:	0.27459107	48	4.819e−11	1.298e−07	TMEM126A:7 NDUFS3:50 NDUFB8:77 NDUFB10:106 TMEM126B:119 NDUFS5:129
Mitochondrial Respiratory Chain Complex	0.27459107	48	4.819e−11	1.298e−07	TMEM126A:7 NDUFS3:50 NDUFB8:77 NDUFB10:106 TMEM126B:119 NDUFS5:129
Proton Motive Force−Driven Mitochondrial	0.25904547	47	8.284e−10	1.487e−06	NDUFS3:50 NDUFB8:77 NDUFB10:106 NDUFS5:129 NDUFA11:137 ATP5PD:194
Oxidative Phosphorylation (GO:0006119)	0.23726723	54	1.684e−09	2.268e−06	NDUFS3:50 NDUFB8:77 NDUFB10:106 NDUFS5:129 NDUFA11:137 ATP5PD:194
Mitochondrial Electron Transport, NADH T	0.28676163	31	3.327e−08	3.584e−05	NDUFS3:50 NDUFB8:77 NDUFB10:106 NDUFS5:129 NDUFA8:422 NDUFA9:482
Proton Motive Force−Driven ATP Synthesis	0.22080975	51	5.025e−08	4.512e−05	NDUFS3:50 NDUFB8:77 NDUFB10:106 NDUFS5:129 NDUFA11:137 ATP5PD:194
Mitochondrial Respiratory Chain Complex	0.17488576	77	1.168e−07	8.992e−05	TMEM126A:7 NDUFS3:50 NDUFB8:77 NDUFB10:106 TMEM126B:119 NDUFS5:129
Aerobic Respiration (GO:0009060)	0.19738478	53	6.821e−07	4.593e−04	NDUFS3:50 NDUFB8:77 NDUFB10:106 NDUFS5:129 NDUFA11:137 NDUFA8:422
Mitochondrial ATP Synthesis Coupled Elec	0.19712403	52	8.968e−07	5.368e−04	NDUFS3:50 NDUFB8:77 NDUFB10:106 NDUFS5:129 NDUFA8:422 NDUFA12:448
Cellular Respiration (GO:0045333)	0.16163905	69	3.530e−06	1.902e−03	NDUFS3:50 NDUFB8:77 NDUFB10:106 NDUFS5:129 NDUFA11:137 NDUFA8:422
Aerobic Electron Transport Chain (GO:001	0.18359846	52	4.741e−06	2.322e−03	NDUFS3:50 NDUFB8:77 NDUFB10:106 NDUFS5:129 NDUFA8:422 NDUFA9:482
Kinetochore Assembly (GO:0051382)	0.41164620	8	5.536e−05	2.485e−02	CENPT:152 CENPC:209 DLGAP5:221 MIS12:559 CENPE:1241 NPC1:1796
piRNA Processing (GO:0034587)	0.25695009	17	2.453e−04	1.016e−01	MOV10L1:33 GPAT2:95 PIWIL1:252 MAEL:444 ASZ1:555 HENMT1:642
Kinetochore Organization (GO:0051383)	0.35007675	9	2.763e−04	1.063e−01	CENPT:152 CENPC:209 DLGAP5:221 MIS12:559 CENPE:1241 NPC1:1796
Positive Regulation Of Macrophage Derive	−0.26742890	15	3.364e−04	1.208e−01	CSF1:36 MAPK9:396 PLA2G5:450 AGTR1:647 CSF2:2070 CD36:2366
tRNA Methylation (GO:0030488)	0.16933665	36	4.413e−04	1.486e−01	TRMT10C:173 THADA:175 TRMT13:300 TRMT10A:710 NSUN2:807 TRMT6:873
Mitochondrial RNA Processing (GO:0000963	0.32164042	9	8.343e−04	2.644e−01	SUPV3L1:54 PNPT1:162 TRMT10C:173 FASTKD1:197 FASTKD5:573 FASTKD2:1355
Positive Regulation Of ERK1 And ERK2 Cas	−0.07948902	141	1.155e−03	3.456e−01	HTR2A:6 CCR7:59 P2RY6:119 NDRG4:233 EGFR:272 CCL21:287
DNA Replication Checkpoint Signaling (GO	0.22840159	15	2.197e−03	4.322e−01	DNA2:114 TIMELESS:352 ORC1:462 CLSPN:1380 TOPBP1:1537 CDC6:1588
Endosomal Transport (GO:0016197)	−0.07465859	145	1.969e−03	4.322e−01	SPAG9:76 CHMP1A:81 STX12:90 VAMP4:165 ERC1:174 REPS2:232
Negative Regulation Of Cytoskeleton Orga	0.18386915	24	1.827e−03	4.322e−01	SHANK3:233 KANK4:456 CLASP1:627 LIMA1:1136 PRKN:1169 CAV3:1665
Peptide Biosynthetic Process (GO:0043043	0.09020074	99	1.963e−03	4.322e−01	MRPL42:66 MRPL37:161 MRPL24:299 WARS1:301 RPS17:348 RPS23:350
Positive Regulation Of Histone H3−K4 Met	0.29357364	9	2.291e−03	4.322e−01	BRCA1:20 KANSL3:685 KAT8:784 HCFC1:1942 SMAD4:2693 PHF20:2784
Positive Regulation Of Transporter Activ	−0.23507867	14	2.327e−03	4.322e−01	PPARD:743 TRPC6:829 PPARGC1A:1238 PON1:1316 SGK2:1789 SYNGR3:2631
Regulation Of Macrophage Derived Foam Ce	−0.17520784	27	1.633e−03	4.322e−01	CSF1:36 MAPK9:396 AGT:439 PLA2G5:450 AGTR1:647 NFKBIA:669
Regulation Of Vesicle Fusion (GO:0031338	−0.37013670	6	1.691e−03	4.322e−01	DOC2B:121 CPLANE2:1076 C2CD5:1838 ANXA1:2227 SPHK1:2655 ANXA2:3876
Regulation Of Vesicle−Mediated Transport	−0.10839144	68	2.021e−03	4.322e−01	YIPF5:7 HIP1:78 RIT2:220 APEX1:778 RIMS2:957 LGI3:999
Regulatory ncRNA Processing (GO:0070918)	0.16747001	28	2.170e−03	4.322e−01	MOV10L1:33 GPAT2:95 PIWIL1:252 MAEL:444 ASZ1:555 HENMT1:642
Translation (GO:0006412)	0.06921375	167	2.097e−03	4.322e−01	MRPL42:66 MRPL37:161 MRPS30:276 PTCD3:278 MRPL24:299 WARS1:301
Granulocyte Activation (GO:0036230)	0.27703198	10	2.419e−03	4.343e−01	CXCL6:94 F2RL1:1237 PRG3:1432 CCL5:1481 IL15:2187 PRKCD:2248
Mitochondrial RNA 3'−End Processing (GO:	0.38978687	5	2.540e−03	4.414e−01	SUPV3L1:54 PNPT1:162 TRMT10C:173 TRNT1:1639 HSD17B10:6281 NA
Regulation Of ERK1 And ERK2 Cascade (GO:	−0.06235331	196	2.713e−03	4.568e−01	HTR2A:6 CCR7:59 P2RY6:119 NDRG4:233 EGFR:272 CCL21:287
Regulation Of Astrocyte Differentiation	−0.30460689	8	2.851e−03	4.654e−01	SERPINE2:406 IL6:858 HES5:1691 CLCF1:2052 F2:2223 EPHA4:2261
Positive Regulation Of Blood Pressure (G	−0.38211424	5	3.086e−03	4.889e−01	TBXA2R:337 GLP1R:1701 CARTPT:1880 SPX:2077 TAC3:2897 NA
T−helper Cell Differentiation (GO:004209	−0.21037951	16	3.580e−03	5.470e−01	FOXP1:234 IL4:356 STAT4:436 BATF:537 IL6:858 GPR183:2083
Lysosome Localization (GO:0032418)	−0.14410533	34	3.656e−03	5.470e−01	VPS33B:32 SPAG9:76 DEF8:472 TFEB:559 PIP4P1:630 RNF167:692
Monoacylglycerol Catabolic Process (GO:0	−0.34013982	6	3.910e−03	5.693e−01	MGLL:5 ABHD12B:503 ABHD16A:1477 ABHD16B:1721 ABHD6:3603 FAAH:7216
Macromolecule Biosynthetic Process (GO:0	0.07521390	122	4.200e−03	5.872e−01	MRPL42:66 MRPL37:161 POLD2:293 MRPL24:299 WARS1:301 RPS17:348
Mitotic Sister Chromatid Segregation (GO	0.08478296	95	4.360e−03	5.872e−01	KIF18B:4 OFD1:76 ZWINT:168 DLGAP5:221 SPAG5:437 SMC4:446
Water−Soluble Vitamin Metabolic Process	−0.13407846	38	4.258e−03	5.872e−01	VNN2:259 MMAA:855 MTHFS:874 SLC19A3:955 BTD:1117 MMADHC:1766

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE	0.31212113	47	1.358e−13	8.808e−10	NDUFS3:50 NDUFB8:77 NDUFB10:106 TMEM126B:119 NDUFS5:129 NDUFAF4:217
REACTOME_COMPLEX_I_BIOGENESIS	0.29026946	48	3.528e−12	1.144e−08	NDUFS3:50 NDUFB8:77 NDUFB10:106 TMEM126B:119 NDUFS5:129 NDUFA11:137
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	0.19765377	79	1.286e−09	2.780e−06	LRPPRC:2 NDUFS3:50 NDUFB8:77 NDUFB10:106 TMEM126B:119 NDUFS5:129
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT_	0.17298816	99	2.812e−09	4.561e−06	LRPPRC:2 NDUFS3:50 NDUFB8:77 NDUFB10:106 TMEM126B:119 NDUFS5:129
WP_OXIDATIVE_PHOSPHORYLATION	0.23375479	46	4.176e−08	5.418e−05	NDUFS3:50 NDUFB8:77 NDUFB10:106 NDUFS5:129 NDUFA11:137 ATP5PD:194
REACTOME_THE_CITRIC_ACID_TCA_CYCLE_AND_R	0.11575033	146	1.430e−06	1.546e−03	LRPPRC:2 NDUFS3:50 NDUFB8:77 NDUFB10:106 TMEM126B:119 NDUFS5:129
KEGG_OXIDATIVE_PHOSPHORYLATION	0.14091874	92	3.050e−06	2.826e−03	ATP6V0A2:16 NDUFS3:50 NDUFB8:77 NDUFB10:106 NDUFS5:129 NDUFA11:137
WP_ELECTRON_TRANSPORT_CHAIN_OXPHOS_SYSTE	0.14199539	72	3.135e−05	2.542e−02	NDUFS3:50 NDUFB8:77 NDUFB10:106 NDUFS5:129 ATP5PD:194 ATP5PB:390
HAMAI_APOPTOSIS_VIA_TRAIL_UP	0.05044349	567	4.503e−05	3.246e−02	BRCA1:20 LRRC40:23 SORL1:25 N4BP2L2:60 PHGDH:78 PLXNC1:96
MOOTHA_VOXPHOS	0.13817574	69	7.292e−05	4.730e−02	NDUFS3:50 NDUFB8:77 NDUFS5:129 ATP5PD:194 CBARP:258 ATP5PB:390
WP_NONALCOHOLIC_FATTY_LIVER_DISEASE	0.10129030	123	1.066e−04	6.287e−02	NDUFS3:50 NDUFB8:77 PIK3R3:86 NDUFB10:106 NDUFS5:129 NDUFA11:137
KEGG_ALZHEIMERS_DISEASE	0.09718728	125	1.783e−04	9.638e−02	NDUFS3:50 NDUFB8:77 NDUFB10:106 NDUFS5:129 ATP5PD:194 ATP2A1:227
KEGG_PARKINSONS_DISEASE	0.11297419	89	2.326e−04	1.160e−01	NDUFS3:50 NDUFB8:77 NDUFB10:106 NDUFS5:129 ATP5PD:194 ATP5PB:390
DAZARD_RESPONSE_TO_UV_NHEK_DN	0.06488606	264	2.956e−04	1.370e−01	AKAP9:26 PER2:121 CEP350:134 MTUS1:200 CENPC:209 RBPJ:225
BIOCARTA_RACC_PATHWAY	0.28587125	13	3.586e−04	1.551e−01	KCNQ3:101 KCNQ5:793 ADCY1:1231 GUCY1A2:1555 NOS3:1563 KCNQ4:2100
PICCALUGA_ANGIOIMMUNOBLASTIC_LYMPHOMA_DN	−0.09675160	107	5.520e−04	2.238e−01	TP53INP2:180 FOXP1:234 CLK1:236 PDE4D:523 SLC30A1:555 SIK3:757
REACTOME_FGFR2_MUTANT_RECEPTOR_ACTIVATIO	−0.23339970	18	6.079e−04	2.320e−01	GTF2F2:168 POLR2F:765 FGF22:1168 FGF8:1546 FGF1:1834 NCBP1:2596
COWLING_MYCN_TARGETS	−0.16206787	36	7.674e−04	2.414e−01	DSEL:42 SHOX2:355 PVALB:482 DDAH2:664 ALPK2:763 SAMD9L:1287
FLORIO_NEOCORTEX_BASAL_RADIAL_GLIA_DN	0.07425692	172	7.973e−04	2.414e−01	KIF18B:4 BRCA1:20 PHGDH:78 HAPLN1:150 POLQ:166 ZWINT:168
CARRILLOREIXACH_MRS3_VS_LOWER_RISK_HEPAT	−0.15314983	39	9.375e−04	2.414e−01	SLC39A5:28 SLC25A1:138 FOXA2:250 SLC29A4:1148 SLC35D2:1430 COASY:2236
KHETCHOUMIAN_TRIM24_TARGETS_UP	−0.14543075	44	8.480e−04	2.414e−01	COL4A1:66 SERPINE1:256 EGR1:423 ICAM1:914 BTG2:1054 ITGAL:1106
GUILLAUMOND_KLF10_TARGETS_UP	−0.14960892	41	9.205e−04	2.414e−01	DAG1:34 SV2A:54 SERPINE2:406 SLC22A7:863 CELSR2:898 DERL1:979
MCBRYAN_PUBERTAL_BREAST_5_6WK_DN	−0.09784915	100	7.302e−04	2.414e−01	PANK3:499 ITSN2:657 DPYSL3:750 VAPB:900 KCNB1:997 PCYT1A:1002
REACTOME_FATTY_ACID_METABOLISM	−0.07692714	155	9.665e−04	2.414e−01	ACSM3:98 SLC25A1:138 TBXAS1:211 GGT1:265 CROT:448 DPEP1:556
DACOSTA_UV_RESPONSE_VIA_ERCC3_DN	0.03588801	752	8.969e−04	2.414e−01	LRPPRC:2 BRCA1:20 AKAP9:26 ZMYND11:39 N4BP2L2:60 PTPN13:128
KAUFFMANN_MELANOMA_RELAPSE_UP	0.13108631	53	9.675e−04	2.414e−01	BRCA1:20 POLQ:166 GMNN:184 PTTG1:310 RAD51AP1:315 DCLRE1A:371
REACTOME_PEPTIDE_LIGAND_BINDING_RECEPTOR	−0.07301325	166	1.194e−03	2.870e−01	CCR7:59 PMCH:141 GALR2:212 CCL21:287 MLNR:326 PRLHR:366
REACTOME_SIGNALING_BY_NUCLEAR_RECEPTORS	−0.06753495	193	1.244e−03	2.881e−01	ZNF217:71 STAG2:113 AXIN1:149 GTF2F2:168 BTC:170 RARB:230
LY_AGING_OLD_UP	−0.34995300	7	1.344e−03	3.005e−01	HTRA1:319 COMP:385 CST6:1116 CRYAB:1293 MMP12:1650 PTGS1:5213
BENPORATH_PRC2_TARGETS	−0.04048309	538	1.409e−03	3.047e−01	FOXL2:10 DDAH1:12 GABRA4:18 BARHL1:91 NOL4:101 HEY1:161
NIKOLSKY_BREAST_CANCER_16P13_AMPLICON	−0.09403078	96	1.468e−03	3.071e−01	AXIN1:149 MPG:278 CCDC78:496 TMEM204:536 MSLNL:599 TSR3:848
NIKOLSKY_BREAST_CANCER_7P22_AMPLICON	0.14713732	38	1.702e−03	3.279e−01	BRAT1:549 MAD1L1:679 AP5Z1:761 SNX8:789 SUN1:850 PAPOLB:930
REACTOME_CARGO_CONCENTRATION_IN_THE_ER	−0.17447757	27	1.703e−03	3.279e−01	SEC24D:23 F8:65 GOSR2:151 LMAN2L:828 CNIH2:949 CTSZ:990
DODD_NASOPHARYNGEAL_CARCINOMA_DN	0.02823659	1111	1.718e−03	3.279e−01	LRPPRC:2 KIF18B:4 ASAP1:11 PARP12:14 BRCA1:20 LRRC40:23
GAUSSMANN_MLL_AF4_FUSION_TARGETS_B_DN	0.31826584	8	1.825e−03	3.354e−01	SH3BP5:903 NIBAN1:912 TRIP12:1377 PLSCR4:1896 WNT5A:2706 ADH7:2994
BIOCARTA_STEM_PATHWAY	−0.24017954	14	1.861e−03	3.354e−01	CSF1:36 IL2:335 IL4:356 IL6:858 CD4:1329 CSF2:2070
LY_AGING_MIDDLE_UP	−0.23780683	14	2.065e−03	3.422e−01	HTRA1:319 COMP:385 CST6:1116 MMP12:1650 DPT:1866.5 CRYBB2:2219
LEE_SP4_THYMOCYTE	−0.25637807	12	2.104e−03	3.422e−01	IRAK2:331 SERPINE2:406 PDE4D:523 IFI44L:568 P2RY1:1085 DUSP4:2780
WP_GABA_RECEPTOR_SIGNALING	−0.16893085	28	1.978e−03	3.422e−01	GABRA4:18 GABRB1:367 ABAT:390 AP2A1:727 GAD1:977 GABRA5:1217
WP_JOUBERT_SYNDROME	0.10633028	70	2.110e−03	3.422e−01	OFD1:76 ARL13B:98 TCTN3:361 CEP290:409 RHOA:673 CEP97:739

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Increased CSF lactate	0.15570833	53	8.928e−05	2.505e−01	LRPPRC:2 NDUFS3:50 TMEM126B:119 NDUFA11:137 TRMT10C:173 NDUFAF4:217
MITOCHONDRIAL COMPLEX I DEFICIENCY	0.22095321	26	9.681e−05	2.505e−01	NDUFS3:50 TMEM126B:119 NDUFA11:137 NDUFAF4:217 TIMMDC1:830 NDUFAF1:1100
Pediatric failure to thrive	0.05256227	474	1.022e−04	2.505e−01	LRPPRC:2 TTC37:3 ATP6V0A2:16 NDUFS3:50 SARS2:85 MRGPRF:103
Undergrowth	0.05396903	465	7.725e−05	2.505e−01	LRPPRC:2 TTC37:3 ATP6V0A2:16 NDUFS3:50 SARS2:85 TMEM126B:119
Failure to gain weight	0.05112044	475	1.552e−04	2.537e−01	LRPPRC:2 TTC37:3 ATP6V0A2:16 NDUFS3:50 SARS2:85 TMEM126B:119
Mitochondrial Diseases	0.06089272	335	1.409e−04	2.537e−01	LRPPRC:2 TMEM126A:7 NDUFS3:50 NDUFB8:77 SARS2:85 NDUFB10:106
NADH:Q(1) Oxidoreductase deficiency	0.21425436	25	2.097e−04	2.571e−01	NDUFS3:50 NDUFB10:106 TMEM126B:119 NDUFA11:137 NDUFAF4:217 TIMMDC1:830
Rhinitis	−0.12707528	72	1.962e−04	2.571e−01	GSTK1:48 TNFRSF1B:131 ADRB2:199 EGFR:272 TBXA2R:337 IL4:356
Abnormal mitochondria in muscle tissue	0.21232253	25	2.392e−04	2.606e−01	NDUFS3:50 NDUFB10:106 TMEM126B:119 NDUFA11:137 NDUFAF4:217 TIMMDC1:830
Acute necrotizing encephalopathy	0.23059768	19	5.032e−04	3.773e−01	NDUFS3:50 TMEM126B:119 NDUFA11:137 NDUFAF4:217 TIMMDC1:830 NDUFAF1:1100
Autosomal Recessive Primary Microcephaly	0.21168551	23	4.426e−04	3.773e−01	STIL:505 OXA1L:570 CDK6:924 MCPH1:1607 CENPJ:1900 PHC1:2011
Ciliopathies	0.07828379	167	5.015e−04	3.773e−01	FAM161A:42 PKHD1:55 OFD1:76 MAPKBP1:90 ARL13B:98 TRAF3IP1:133
Progressive macrocephaly	0.21476168	22	4.901e−04	3.773e−01	NDUFS3:50 TMEM126B:119 NDUFA11:137 NDUFAF4:217 TIMMDC1:830 NDUFAF1:1100
Walker−Warburg congenital muscular dystr	−0.23565765	18	5.387e−04	3.773e−01	DAG1:34 COL4A1:66 POMT1:193 RXYLT1:514 LARGE1:679 POMK:1335
Increased hepatocellular lipid droplets	0.32831553	9	6.483e−04	4.238e−01	LRPPRC:2 COA7:435 COA5:899 FASTKD2:1355 COX20:2518 SCO1:3493
Non−small cell carcinoma	−0.28133909	12	7.402e−04	4.536e−01	EGFR:272 NNAT:1138 ID1:1206 GRIN2B:1403 FHIT:1602 ESR2:1611
Diaphragmatic Hernia	−0.13146551	53	9.396e−04	5.118e−01	RARB:230 FOXA2:250 THRB:271 AGT:439 LRP1:512 AGTR1:647
Metatarsal Valgus	−0.23917061	16	9.273e−04	5.118e−01	DAG1:34 COL4A1:66 POMT1:193 RXYLT1:514 LARGE1:679 SIL1:899
Low Grade Prostatic Intraepithelial Neop	−0.33607952	8	9.960e−04	5.140e−01	EGFR:272 SPP1:1806 ERG:2155 BCL2:2156 TSPAN13:2200 NPY:2914
Coronary Arteriosclerosis	−0.03887490	616	1.131e−03	5.306e−01	HTR2A:6 SELP:8 DDAH1:12 PRKAA1:41 GSTK1:48 CCR7:59
Ganglioneuroblastoma	−0.15952475	34	1.293e−03	5.306e−01	EGFR:272 KDM1A:293 MYCN:596 HIC1:1313 GAP43:1386 FGF8:1546
Increased serum lactate	0.09598970	95	1.246e−03	5.306e−01	LRPPRC:2 NDUFS3:50 TRMT10C:173 SFXN4:187 TMEM70:333 RARS2:335
Muscle eye brain disease	−0.23143014	16	1.353e−03	5.306e−01	DAG1:34 COL4A1:66 POMT1:193 RXYLT1:514 LARGE1:679 POMK:1335
Renal cortical microcysts	0.41455602	5	1.326e−03	5.306e−01	TTC37:3 PEX1:865 NEK8:891 PEX2:1068 INVS:3670 NA
Rheumatoid Arthritis	−0.02667427	1379	1.242e−03	5.306e−01	HTR2A:6 SELP:8 DDAH1:12 CSF1:36 GSTK1:48 CD40LG:68
Polyarticular Juvenile Idiopathic Arthri	−0.22364473	17	1.413e−03	5.330e−01	STAT4:436 LTBR:437 IL6R:1173 IL2RB:1754 TYK2:2210 ZFP36L1:2750
Abnormal aldolase level	−0.22668882	15	2.371e−03	5.899e−01	DAG1:34 COL4A1:66 POMT1:193 RXYLT1:514 LARGE1:679 SIL1:899
Abnormal levels of creatine kinase in bl	−0.21924578	16	2.399e−03	5.899e−01	DAG1:34 COL4A1:66 POMT1:193 RXYLT1:514 LARGE1:679 SIL1:899
Absence Seizures	−0.08261979	111	2.688e−03	5.899e−01	SIGMAR1:47 GAMT:63 ACHE:64 GOSR2:151 SCN1A:181 ABAT:390
Agammaglobulinemia	−0.17147797	25	3.008e−03	5.899e−01	CD40LG:68 IL2RG:641 TNFRSF11A:672 IL6:858 SLC3A2:1289 SH2D1A:1392
Carcinoma ex pleomorphic adenoma	−0.22361573	15	2.716e−03	5.899e−01	EGFR:272 IL6:858 NCOR1:1013 CCND1:1728 PLAG1:2168 TYK2:2210
Cardiovascular Diseases	−0.03644723	655	1.677e−03	5.899e−01	HTR2A:6 SELP:8 DDAH1:12 CSF1:36 COL5A2:38 PRKAA1:41
Coronary heart disease	−0.03347632	742	2.214e−03	5.899e−01	SELP:8 DDAH1:12 PRKAA1:41 GSTK1:48 CCR7:59 TBX20:60
Dysdiadochokinesis	0.15430429	31	2.957e−03	5.899e−01	XRCC4:365 FMR1:547 PLA2G6:758 SETX:1556 GRM1:1633 ELOVL4:1727
Fibromyalgia	−0.13503961	41	2.789e−03	5.899e−01	HTR2A:6 PRKAA1:41 ADRB2:199 TCP1:325 IL4:356 SCLY:722
Fibrosis	−0.07216252	144	2.867e−03	5.899e−01	P2RY6:119 ADRB2:199 KIF21A:252 SERPINE1:256 EGFR:272 IL4:356
Frontal lobe hypoplasia	0.19791939	21	1.695e−03	5.899e−01	STIL:505 CDK6:924 MCPH1:1607 FIG4:1740 CENPJ:1900 BRAF:1975
Generalized glycogen storage disease of	0.21006030	17	2.717e−03	5.899e−01	UGT1A6:163 FUT1:766 TNNT2:1272 GLA:1603 IGF2R:1770 G6PC:2067
Hypoplastic labia minora	0.22525750	16	1.814e−03	5.899e−01	GMNN:184 SETBP1:236 ORC1:462 DVL3:870 ORC4:1474 CDC6:1588
Impaired exercise tolerance	0.10809566	67	2.240e−03	5.899e−01	NDUFS3:50 DNA2:114 TMEM126B:119 NDUFA11:137 SLC18A3:213 NDUFAF4:217

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0000372 irregular coat pigmentation	−0.26245980	14	6.818e−04	3.239e−01	TRAPPC6A:246 MCOLN3:343 TBX15:500 OCA2:1088 TYR:1276 HTR2C:1772
MP0005380 embryogenesis phenotype	0.03723377	373	1.654e−02	6.548e−01	BRCA1:20 SUPV3L1:54 GMNN:184 RBPJ:225 RRN3:295 PTPRB:325
MP0010234 abnormal vibrissa follicle	0.37079936	4	1.024e−02	6.548e−01	ST14:560 CHUK:785 PKP3:2118 SPINK5:4842 NA NA
MP0003077 abnormal cell cycle	0.06418372	139	9.748e−03	6.548e−01	BRCA1:20 CTNND1:183 GMNN:184 PTTG1:310 LATS2:366 FANCM:380
MP0002136 abnormal kidney physiology	0.04359925	267	1.634e−02	6.548e−01	NT5E:43 TRPV6:62 ITGA3:261 PTTG1:310 SCNN1G:383 EPHB6:452
MP0001485 abnormal pinna reflex	0.15536536	29	3.857e−03	6.548e−01	PTPRQ:154 OTOG:312 FBXO11:706 SLC26A4:921 SLC12A7:1506 MYC:1558
MP0000858 altered metastatic potential	−0.09971146	55	1.085e−02	6.548e−01	BHLHA15:354 CTSZ:990 ITGAL:1106 MTBP:1306 HIC1:1313 MFGE8:1799
MP0005501 abnormal skin physiology	0.08472839	68	1.623e−02	6.548e−01	DGKD:536 ST14:560 NSMAF:604 CHUK:785 IFNG:1044 F2RL1:1237
MP0010771 integument phenotype	0.05577369	175	1.205e−02	6.548e−01	SUPV3L1:54 PPIB:97 DGKD:536 ST14:560 NFKBIZ:594 NSMAF:604
MP0001216 abnormal epidermal layer	0.05222954	201	1.197e−02	6.548e−01	BRCA1:20 SUPV3L1:54 ZDHHC13:102 CTNND1:183 RBPJ:225 ITGA3:261
MP0001672 abnormal embryogenesis/ devel	0.03723377	373	1.654e−02	6.548e−01	BRCA1:20 SUPV3L1:54 GMNN:184 RBPJ:225 RRN3:295 PTPRB:325
MP0005025 abnormal response to	−0.04437662	339	6.274e−03	6.548e−01	SELP:8 CD40LG:68 TNFRSF1B:131 SERPINE1:256 IL4:356 ANTXR2:365
MP0001765 abnormal ion homeostasis	0.05975646	99	4.144e−02	7.242e−01	TRPV6:62 PLCL2:241 SCNN1G:383 EPHB6:452 SLC4A1:688 SLC2A9:736
MP0005248 abnormal Harderian gland	−0.17776734	14	2.141e−02	7.242e−01	RARB:230 VSX2:257 ALDH1A1:349 ALDH1A3:2906 RARA:3082 RXRA:4396
MP0004133 heterotaxia	−0.10177227	30	5.422e−02	7.242e−01	CITED2:70 FOXA2:250 VANGL1:358 MGAT1:399 DLL1:915 PITX2:1177
MP0003984 embryonic growth retardation	0.03240778	302	5.858e−02	7.242e−01	BRCA1:20 RRN3:295 PTPRB:325 LATS2:366 HIRA:430 CREBBP:439
MP0002166 altered tumor susceptibility	−0.11942817	22	5.286e−02	7.242e−01	KLF4:815 PDCD4:1411 PSME1:1604 RAG1:1708 IRF4:2394 IFNGR1:2808
MP0002332 abnormal exercise endurance	−0.10231125	29	5.705e−02	7.242e−01	ACHE:64 ADRB2:199 SCO2:352 IL6:858 CAV2:1009 APLN:1822
MP0009672 abnormal birth weight	0.08386376	53	3.541e−02	7.242e−01	GHR:38 DGKD:536 LAMA1:709 DNMT3B:1015 KIF1A:1083 HNRNPD:1296
MP0000689 abnormal spleen morphology	−0.02938272	510	2.934e−02	7.242e−01	SELP:8 PRKAA1:41 CCR7:59 CD40LG:68 CITED2:70 KLHL6:72
MP0004272 abnormal basement membrane	−0.09957551	29	6.402e−02	7.242e−01	COL4A1:66 ILK:147 FLRT3:938 CAV2:1009 HSPG2:1115 PLOD3:1331
MP0010030 abnormal orbit morphology	−0.16029343	12	5.473e−02	7.242e−01	RARB:230 PITX2:1177 ROR2:2025 SMO:2049 OTX1:4411 LMNA:4465
MP0001849 ear inflammation	0.14025387	21	2.631e−02	7.242e−01	MECOM:424 FBXO11:706 EYA4:976 ITGB4:2073 PLCG1:3003 PSTPIP2:3016
MP0003724 increased susceptibility to	−0.13769098	15	6.513e−02	7.242e−01	SELP:8 CD200:654 PPARG:2328 TNFSF10:2360 ZAP70:2747 IFNGR1:2808
MP0003045 fibrosis	0.11413073	22	6.429e−02	7.242e−01	PKHD1:55 ENTPD1:142 ST14:560 NOS1:1130 SH2B3:2023 DST:2142
MP0009840 abnormal foam cell	0.16565295	11	5.731e−02	7.242e−01	SUPV3L1:54 SMPD1:1119 NPC1:1796 ABCG1:1801 APOE:2126 PLA2G15:2944
MP0003880 abnormal central pattern	−0.20489747	10	2.496e−02	7.242e−01	NLGN1:1098 DBX1:1985 DSCAM:2238 EPHA4:2261 NCK1:4256 EFNB3:4273
MP0001756 abnormal urination	0.06237639	79	5.675e−02	7.242e−01	TRPV6:62 PTTG1:310 SCNN1G:383 EPHB6:452 SLC4A1:688 UPK2:697
MP0002088 abnormal embryonic growth/wei	0.03539614	308	3.709e−02	7.242e−01	BRCA1:20 RBPJ:225 RRN3:295 PTPRB:325 LATS2:366 HIRA:430
MP0005164 abnormal response to	−0.06392640	72	6.213e−02	7.242e−01	SIRT7:96 SERPINE2:406 EGR1:423 CLOCK:495 NGFR:520 IRF1:643
MP0002060 abnormal skin morphology	0.07285997	72	3.350e−02	7.242e−01	SUPV3L1:54 PPIB:97 NFKBIZ:594 CHUK:785 COL3A1:863 DKC1:969
MP0002067 abnormal sensory capabilities	−0.03510747	238	6.708e−02	7.242e−01	SELP:8 ACHE:64 LMX1A:124 CALB1:148 SCN1A:181 FOXA2:250
MP0000367 abnormal coat/ hair	0.03524761	297	4.125e−02	7.242e−01	BRCA1:20 ADAMTS20:28 SUPV3L1:54 TRPV6:62 ZDHHC13:102 LPIN1:111
MP0002882 abnormal neuron morphology	−0.02297325	813	3.775e−02	7.242e−01	DAG1:34 CSF1:36 BARHL1:91 FBXO2:117 TNFRSF1B:131 ABCD2:133
MP0003011 delayed dark adaptation	−0.16778817	10	6.635e−02	7.242e−01	PLIN2:1305 RBP1:2329 RDH11:3072 RDH5:3411 RLBP1:3440 RXRA:4396
MP0002796 impaired skin barrier	0.11078314	36	2.178e−02	7.242e−01	ST14:560 NSMAF:604 CHUK:785 TGM1:1644 ELOVL4:1727 RARG:1734
MP0002653 abnormal ependyma morphology	−0.21667109	9	2.448e−02	7.242e−01	DLG5:254 VAX1:1200 NR1H2:3008 E2F5:3835 BBS1:4810 HYDIN:4813
MP0005395 other phenotype	0.06187173	83	5.283e−02	7.242e−01	PKHD1:55 ST3GAL1:115 ENTPD1:142 MACF1:254 SCNN1G:383 ZAR1:450
MP0009785 altered susceptibility to	−0.07595530	67	3.243e−02	7.242e−01	ANTXR2:365 CXCR3:378 STAT4:436 LTBR:437 C4B:449 NFKBIA:669
MP0005410 abnormal fertilization	0.08227690	58	3.094e−02	7.242e−01	DKKL1:208 ADAD1:239 CATSPER2:1145 SLC26A8:1396 INPP5B:1464 ROS1:1513

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
hippocampus.proper	−0.30801276	9	1.381e−03	7.457e−02	CNIH2:949 OLIG2:1250 FEZF2:1431 NTS:1681 SLC17A7:2150 GRP:2533
testis	0.02631645	1339	2.779e−03	7.503e−02	FBXO24:10 CATSPERE:13 MOV10L1:33 RANBP17:41 RIBC1:46 ZSCAN5B:56
EBV.lymphocyte	0.03275550	652	5.860e−03	1.055e−01	KIF18B:4 BRCA1:20 SORL1:25 CHAC2:63 SLFN14:89 SCIMP:92
cerebellar.hemisphere	−0.03676513	459	8.409e−03	1.135e−01	SV2A:54 BARHL1:91 NOL4:101 ABCD2:133 TAFA3:182 RIT2:220
coronary.artery	−0.16795084	14	2.967e−02	3.204e−01	COL4A1:66 GJA5:404 CCDC190:851 PCOLCE2:1463 HEYL:2874 ST6GALNAC5:3213
bladder	0.10177576	23	9.144e−02	3.814e−01	UGT1A6:163 PADI3:317 UPK2:697 SNX31:741 PSCA:1104 MSX2:2390
Brodmann.area.9	−0.04063624	162	7.669e−02	3.814e−01	HTR2A:6 AK5:116 CADM2:125 SCN1A:181 DACH2:276 NWD2:517
cortex.kidney	0.04572348	110	9.942e−02	3.814e−01	PKHD1:55 NPHS1:484 CLRN3:613 GCGR:727 SLC22A8:816 AQP6:884
greater.omentum	−0.09772686	32	5.608e−02	3.814e−01	RBP7:217 ADAMTS4:312 IL6:858 BTNL9:1221 AADACL4:2198 ALOX15:2367
nucleus.accumbens	−0.04926243	93	1.022e−01	3.814e−01	GABRA4:18 SH2D5:105 RGS20:187 DACH2:276 GABRB1:367 RGS8:563
sigmoid.colon	−0.06969206	50	8.900e−02	3.814e−01	BTC:170 GALR2:212 NKX2−3:1432 CHRNA3:1508 CHRM2:1582 FOXP2:1828
skeletal.muscle	0.03259972	232	9.093e−02	3.814e−01	NEXN:24 IGFN1:192 CAVIN4:202 ATP2A1:227 NEB:264 MYBPC2:441
spleen	−0.02976609	291	8.525e−02	3.814e−01	CD40LG:68 KLHL6:72 PRR5:99 P2RY6:119 S100Z:129 RENBP:166
tibial.artery	0.04837169	101	9.464e−02	3.814e−01	TBX18:164 SYTL2:273 ROR1:413 CHSY3:464 LMOD1:588 CPED1:831
transformed.skin.fibroblast	−0.03048819	243	1.059e−01	3.814e−01	SH3BP4:4 YIPF5:7 ADAMTS2:17 SEC24D:23 EVC:25 CSF1:36
endocervix	0.05331757	70	1.242e−01	4.159e−01	ANGPT4:607 SPON2:630 COL24A1:820 ADRA2C:883 ADAM33:893 RGPD2:991
tibial.nerve	−0.03203464	183	1.386e−01	4.159e−01	ANGPTL7:40 VIT:134 TMPRSS5:136 DKK2:186 ADAMTS14:216 TLDC2:364
eye.development	−0.05770067	57	1.329e−01	4.159e−01	ALDH3A1:57 CITED2:70 VSX2:257 KLF4:815 PITX2:1177 VAX1:1200
amygdala	−0.13649509	8	1.814e−01	4.897e−01	SLC17A7:2150 TNR:3202 PTPRZ1:3730 BCAN:4763 NEUROD2:4888 NEUROD6:4894
Brodmann.area.24	−0.07276936	29	1.756e−01	4.897e−01	DOK6:167 DRD5:403 GAD1:977 SLC6A1:1791 PTER:1948 SLC17A7:2150
left.ventricle	0.04363743	67	2.183e−01	5.612e−01	TNNI3K:125 MYPN:481 CKMT2:524 HSPB3:541 SMPX:719 TNNT2:1272
transverse.colon	0.02840343	144	2.428e−01	5.959e−01	SMPDL3B:73 VIL1:212 OMP:279 NR1I2:285 MOGAT2:289 UCN3:311
pituitary.gland	−0.02296925	202	2.653e−01	6.229e−01	JAKMIP3:30 LMX1A:124 B9D1:126 COL11A2:197 CHRNE:281 KY:314
aorta	0.02394348	151	3.135e−01	6.240e−01	ST6GAL2:18 VCAN:51 NAV1:105 HAPLN1:150 LTBP2:329 FILIP1L:388
cerebellum	−0.01570143	354	3.190e−01	6.240e−01	KCNAB3:11 ACHE:64 CORO2B:86 BARHL1:91 DOC2B:121 CALB1:148
esophagogastric.junction	−0.10082003	8	3.236e−01	6.240e−01	COL4A5:2445 NKX6−1:4185 F2RL2:4692 CHRM3:6134 ADCY5:6986 GADL1:7512
esophagus.mucosa	0.02009126	213	3.174e−01	6.240e−01	EPS8L1:48 UGT1A6:163 KLK13:201 DENND2C:226 PADI3:317 CDCP1:718
prostate	−0.04940382	37	2.992e−01	6.240e−01	ACP3:513 SERPINB11:712 CPA6:1133 MKX:1558 ARHGEF38:1597 EPHA3:1927
subcutaneous.adipose	−0.03210751	71	3.512e−01	6.540e−01	AGTR1:647 LHX6:697 LGALS12:865 AVPR2:928 SLC19A3:955 OPN4:1058
blood	−0.01585344	266	3.799e−01	6.839e−01	TREML2:20 TNFRSF1B:131 BCL2A1:198 TBXAS1:211 EFHD2:235 VNN2:259
adrenal.gland	−0.02024691	107	4.717e−01	7.718e−01	CYP21A2:311 MCOLN3:343 C4B:449 AGRP:454 ALAS1:825 IDH1:835
caudate.nucleus	−0.04144314	26	4.650e−01	7.718e−01	MAPK4:524 SLC35D3:1053 VAX1:1200 FAM237A:1421 PSD2:1856 SLC1A2:1971
ovary	−0.02167629	98	4.606e−01	7.718e−01	FOXL2:10 ACSM3:98 PLA2G5:450 TLR5:842 CDK15:1227 LAG3:1262
fallopian.tube	−0.02645210	48	5.270e−01	8.370e−01	EGR1:423 MSLNL:599 STOML3:862 BRS3:986 GASK1A:1055 CFAP157:1063
cerebral.cortex	−0.01620670	103	5.718e−01	8.519e−01	KCNAB2:26 NWD2:517 RASL10A:678 KCNK4:736 PHF24:787 SEMA5B:839
hypothalamus	−0.01913363	58	6.152e−01	8.519e−01	PMCH:141 ENKUR:357 BRINP3:1210 SNCG:1267 NTS:1681 CARTPT:1880
lung	0.01311892	139	5.959e−01	8.519e−01	DNASE2B:123 RTKN2:171 BPIFA1:238 ADRB1:271 PTPRB:325 COL6A5:330
minor.salivary.gland	0.01794035	69	6.075e−01	8.519e−01	CRACR2B:75 DNASE2B:123 MMP27:403 ENTPD3:495 BPIFB1:1404 KCNN4:1795
uterus	0.02179657	53	5.840e−01	8.519e−01	MRGPRF:103 ADAMTS19:148 ATP2B4:155 HOXD4:234 FBXL22:1683 KCNG1:1971
vagina	−0.01458443	89	6.359e−01	8.584e−01	TMPRSS11E:493 FAM83C:587 SBSN:645 SPINK7:671 SERPINB11:712 GJB5:846